Falsifiable Predictions of the Rope Hypothesis

Fifteen Predictions and a Reproducibility Standard

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Abstract

This note collects the falsifiable predictions of the rope hypothesis in one place, for outside review. The hypothesis is a non-standard, self-published framework; nothing here asks the reader to accept its ontology. Each prediction is a stated mathematical relation with its inputs classified as derived or measured, its test named, and its falsifier spelled out. The inventory now stands at fifteen: four testable numbers (Part I), two structural consequences (Part II), three predictions from strand extensibility and molecular geometry (Part III), and the six-member measurement-sector family added at v2.2.5 from the strand-level campaign (Part IV), plus one declared retrodiction. This document supersedes all earlier prediction notes; it is the single inventory, maintained under one standard.

1. How to Read This Note

A prediction is only as good as its honesty about what stands behind it. For each prediction below we separate three things a reviewer needs:

The number or shape — the specific quantity or structure the framework commits to, with no remaining freedom to adjust.

The inputs — stated explicitly as DERIVED (forced by the rope structure) or MEASURED (taken from experiment as a fixed input). A prediction that leans on a measured input is still falsifiable, but it is not parameter-free, and we say so.

The test and the falsifier — the experiment that will decide, and the result that would kill the prediction.

None of these predictions depends on accepting the rope ontology. Each follows from a stated mathematical relation; a reader may accept, test, or reject any one independently of the others and of the framework that motivated it. One caveat governs several entries in Part IV: the framework has no derived absolute scale, so those entries commit to SHAPES and RELATIONS while leaving locations in frequency or length open; each such entry is marked scale-open.

2. The Predictions at a Glance

#	Prediction	Committed number / shape	Test (arena)	Status / horizon
1	MOND acceleration scale $g†$ = $cH_0/2\pi$	1.04×10^{-10} m/s^2	H_0 convergence; z-drift of $g†$	DESI/Euclid, ongoing
2	Neutrino mass sum $\Sigma m\nu$	58.5 meV	CMB + BAO + galaxy surveys	CMB-S4/Euclid, ~2027+
3	Cosmic birefringence EB/EE	0.0119, flat in ℓ	CMB EB cross-power	LiteBIRD, ~2030s
4	MOND interpolation shape	(retired)	Galaxy rotation curves	RETIRED at GRV-031
5	Gauge-boson Koide structure	structural	(no direct test yet)	—
6	α–G covariation, d ln α = −2 d ln G	ratio −2 (provisional)	quasar $\dot{\alpha}$ + LLR/pulsar $\dot{G}$ bounds	available now (bounds)
7	Vacuum Kerr nonlinearity	3:1 anisotropy, negative Δn	vacuum birefringence (PVLAS-class)	confronted; window open
8	Dark longitudinal tension channel	superluminal, cubic-only coupling	stress-channel searches	structural
9	Heavy-hydride 90° asymptote	H_2Po 90.0–90.7°; BiH_3 90.0–91.5°	molecular spectroscopy	available now
10	Gapped weave noise floor	gap + band $\times\sqrt{(1+4kt)}$; no soft modes	decoherence spectroscopy	scale-open
11	Detection = nucleation kinetics	first-passage package	threshold-detector statistics	available now
12	Spin-½ requires extension	substructure at some scale	compositeness searches	one-sided
13	CHSH cap, failed organ named	$2\sqrt{2}$ = 2.828	loophole-free Bell tests	standing
14	Born exponent from channel energy	p = 2	threshold response calibration	available now
15	Analog universality (cliff + coverage)	slope ≈ −8.5/width; crossover at contact size	pendulum/cold-atom/domain-wall analogs; filament brushes	available now

Entries 5 and 8 are structural consequences stated for completeness rather than near-term numbers; entry 6 is provisional on one derivation, as its section states; scale-open entries commit *to shape while the absolute scale remains underived; entry 12 is one-sided in practice. Entries 10–15 were added at v2.2.5 from the strand-level measurement campaign.*

Part I — Four Testable Numbers

Prediction 1 — The MOND Acceleration Scale

The relation. The MOND characteristic acceleration $g† \approx 1.2\times10^{-10}$ m/s² is an empirical parameter in Modified Newtonian Dynamics and in the Radial Acceleration Relation observed in disk galaxies. The rope framework proposes it is not free but fixed by the Hubble rate:

$$g† = c H_0 / 2\pi = c^2 / (2\pi R\_H)$$

connecting the galactic acceleration scale directly to cosmology using only c, H_0, and a geometric factor 2π. No galaxy-specific fitting enters.

Inputs. MEASURED: H_0 (the Hubble constant); the prediction inherits the Hubble tension — an irreducible ~9% spread (H_0 = 67–73 km/s/Mpc) until H_0 is pinned. DERIVED (partially): the form $cH_0/2\pi$ from the horizon coupling. HONEST CAVEAT: the 2π is motivated by horizon-sphere geometry but is not uniquely forced; the alternative $cH_0/6$ performs comparably against SPARC (RMS 0.203 vs 0.201). We do not claim the coefficient is settled.

Test and falsifier. With current local H_0, $g† = cH_0/2\pi$ sits about 6% below the SPARC value — right ballpark, not confirmed. The sharper, framework-specific test is drift: g† and H_0 should track together across cosmic time. A measurement of g† from high-redshift galaxy samples that does NOT scale with H(z) would falsify the relation; so would H_0 converging far from the value that makes the relation hold, with no z-drift compensating.

Prediction 2 — The Sum of Neutrino Masses

The relation. The Koide formula relates the three charged-lepton mass amplitudes to a phase on a circle; the measured lepton masses fix that phase at φ = 132.733°. Applying the same Koide structure to neutrinos with a fixed offset π/12 = 15° (Brannen 2006) gives φ_ν = 147.733°. With the measured solar splitting as the one scale input, all three neutrino masses are then determined:

$$\Sigma m_\nu = 58.5 \text{ meV}$$

Inputs. MEASURED: the solar mass splitting Δm^2_{21} (the one scale), and the ratio $\Delta m^2_{31}/\Delta m^2_{21}$ = 32.6 used to fix the offset. INPUT, NOT DERIVED: the π/12 offset itself. HONEST CAVEAT: π/12 is an empirical choice, not derived from the rope structure. A skeptic who treats the offset as fitted will regard Σm_ν as a consistency check rather than a pure prediction. We state it as the former.

Test and falsifier. Σm_ν = 58.5 meV is the one number here not yet measured, with nothing left to adjust. It sits just above the normal-ordering minimum (~59 meV), making it sharp and risky: distinguishable both from the quasi-degenerate spectrum (Σ ~ 100–200 meV) and, with care, from the bare splitting minimum. Falsified by a confirmed Σm_ν below ~50 meV or above

~70 meV, or by confirmation of inverted ordering. Cosmological bounds are already pressing toward this range.

Prediction 3 — Cosmic Birefringence (EB/EE)

The relation. The rope is a chiral object: its two strands wind with a definite handedness. That chirality acts on propagating light as a Chern-Simons-type coupling, rotating the plane of linear polarization as it travels — cosmic birefringence. Integrated to the last-scattering surface, the rotation angle is $\beta = \chi cD/2$ with $\chi = \sin(2\theta\_W)$ n_rope r_H^2/c. Eskilt & Komatsu (2022) report $\beta = 0.342° \pm 0.094°$ from Planck; matching that fixes the rope chirality product to $\sin(2\theta\_W)$ n_rope $r\_H^2 = 9.18\times10^{-29}$ m^{-1}. The framework then predicts the CMB EB cross-correlation:

EB/EE = sin(4β)/2 = 0.0119, flat in multipole ℓ

Inputs. DERIVED: the mechanism (a parity-odd CS coupling from the rope helix) and the functional form EB/EE = sin(4β)/2, flat in ℓ — the flatness is a genuine rope-specific signature; many alternative mechanisms predict ℓ-dependence. MEASURED: the rotation angle β (from Planck), which fixes the chirality product; the individual geometric factors are not separately determined.

Test and falsifier. LiteBIRD (launch ~2030s) is designed to measure CMB EB to the precision this prediction requires. Falsified if EB/EE is inconsistent with sin(4β)/2 at the measured β, or if the cross-correlation shows significant multipole dependence rather than the predicted flatness. This is the cleanest near-term test of a rope-specific mechanism.

Prediction 4 — The MOND Interpolation Shape [RETIRED at GRV-031]

RETIRED at GRV-031 (July 2026), kept visible per house rules. The original claim -- a tanh-type interpolation shape differing ~8 percent from standard forms -- was never pinned to an exact nu(y) (the audit finding of GRV-030), and the corpus-natural derivation candidate (Langevin alignment-saturation against the horizon floor, parameter-free) was stated, bar-locked, and tested on 2788 SPARC points: it loses to the simple interpolation by 0.015 dex at the predicted g_dagger. Both the tanh family and the derived candidate are disfavored; the shape prediction is retired. The framework's registered MOND content is Prediction 1 -- the acceleration scale, confirmed at zero free parameters on the same dataset -- carried by the horizon channel alone (no galaxy-scale scalar exists; GRV-028/029). Any future shape candidate must beat 0.1421 + 0.003 dex at the predicted g_dagger to enter this paper.

Part II — Structural Consequences

Prediction 5 (Structural) — Gauge-Boson Koide Structure

At the rope's Chern-Simons level k = 3, the quantum dimensions of the spin-½ and spin-1 representations coincide ($d\_{1/2} = d_1 = \Phi$, the golden ratio, because $\sin(2\pi/5) = \sin(3\pi/5)$). This forces the W and Z bosons onto the SAME phase structure $\varphi = (3+\Phi)\theta\_W$ that governs the charged leptons — a lepton/gauge-boson link the Standard Model does not contain. The underlying identity is exact mathematics. Stated as structural only: a clean consequence without a direct experimental test at present, and not counted among the four numbers of Part I.

Prediction 6 (Structural / Provisional) — Correlated Variation of α and G

The relation. The electromagnetic and gravitational couplings are expressed in the SAME rope-medium primitives: $\alpha \sim 2T^2/(\kappa a)$ from the EM energy coefficient, and G governed by the same tension T. Because both are functions of one shared set of primitives, α and G are NOT independent — a sharp departure from standard physics. If a shared primitive drifts, they co-vary in a fixed, scale-free ratio; for tension-driven drift:

$$\mathbf{d\ln\alpha = -2\,d\ln G \quad (equivalently\ \dot{\alpha}/\alpha = -2\,\dot{G}/G)}$$

Inputs. DERIVED: $\alpha \sim 2T^2/(\kappa a)$, whose chain is derived to rope-medium primitives with the locking energy $J = T^2/\kappa$ exact in the harmonic regime. ASSUMED: $G \sim 1/(Ta)$, the natural tension-rigidity scaling, NOT yet derived to the same rigor — the specific exponent −2 depends on it.

Test and falsifier. Testable now against combined bounds: quasar-absorption limits on $\dot{\alpha}$ and lunar-laser-ranging / pulsar-timing limits on $\dot{G}$. A measured drift ratio inconsistent with −2 (for tension-channel drift) falsifies the shared-tension structure. Two qualifications: (i) the −2 is PROVISIONAL on deriving $G(T,\kappa,a)$ rigorously — the framework-forced content is a FIXED ratio, not the specific value; (ii) current data are consistent with no variation of either constant, so this is a bound and a standing prediction, not a detection.

Part III — Extensibility and Molecular Geometry

Prediction 7 — Vacuum Kerr Nonlinearity, Quantified and Confronted

The relation. The strand stretch modulus k, required for the repulsive core (EM-RECON-009), makes light self-interact at quartic order with onset strain $g^* = \sqrt{(4T_0/(k-T_0))}$ (EM-RECON-010). The polarization structure is computed exactly: anisotropy ratio 3:1 with NEGATIVE index shifts (background fields stiffen the network; light speeds up), versus QED's 7:4 with positive shifts — a double qualitative discriminator (EM-RECON-016).

Confrontation to date. Matching the magnitude against the final PVLAS limit EXCLUDES identifying the rope quartic with the observed light-by-light nonlinearity (overshoot ~570×) and sets $\Sigma \gtrsim 8.6\times10^{27}(k/T_0-1)\ \mathrm{J/m^3}$ — the framework has already been corrected once by this confrontation, and says so. Decisive window: vacuum-birefringence experiments improving toward QED sensitivity will either detect a negative Δn with 3:1 anisotropy (rope signature) or push Σ higher and confirm QED. Either outcome is informative.

Prediction 8 (Structural) — A Dark Longitudinal Tension Channel

Stability requires $k > T_0$, so longitudinal strand waves propagate at $c\_L\text{,eff} = \sqrt{((k+\lambda\gamma^2\tau_0^2)/\mu)} > c$, with $\gamma = 1/\sin^2\theta$ from helix geometry (EM-RECON-012). The channel decouples from matter and light at linear order exactly (cubic-only coupling) and creates no causal paradox in the emergent-Lorentz framework (EM-RECON-011). Any observed linear coupling to a superluminal stress channel, or its firm exclusion at the predicted nonlinear coupling strength, bears directly on the extensibility primitive.

Prediction 9 — The Heavy-Hydride 90-Degree Asymptote

The relation. The molecular-geometry theorems make 90 degrees the raw fixed point for single bonds on orthogonal dipolar lobes, approached from above as the positive opening corrections collapse down each column. Confirmed on eight hydrides (H_2O 104.48° → H_2Te 90.25°; NH_3 106.67° → SbH_3 91.70°): monotone, one-sided (a single 89-degree hydride would have falsified the structure), with the $XH_3 > XH_2$ crowding law holding at every period.

Standing predictions, registered in advance: H_2Po between 90.0° and 90.7°; BiH_3 between 90.0° and 91.5° — one-sided windows. Tier label: consistency-tier against quantum chemistry (which predicts these angles too); the distinctive content is the asymptote's derivation from two integration-verified theorems rather than a hybridization narrative.

Part IV — The Measurement-Sector Family (added at v2.2.5)

With the v2.2.5 cycle, the measurement chain — amplitude, detection event, reservoir, and reservoir spectrum — was constructed on literal strands (FND-STRAND-005 through 008), and a family of falsifiable commitments follows. Scale-open entries commit to shapes and relations; the locations remain underived.

Prediction 10 — The Gapped Environmental Noise Floor (scale-open)

The relation. The weave reservoir's normal-mode spectrum is gapped — every mode carries the strand mass, band [ω_{gap}, $\omega_{gap}\sqrt{(1+4kt)}$], NO soft modes; environmental noise sourced by the fundamental medium arrives only at and above the gap. DERIVED: the gap's existence and the band shape, from the engine's own Hessian (FND-STRAND-008; matched to dispersion at 7×10^{-16}). OPEN: the gap's location in hertz. Falsifier: an established, irreducible, gapless environmental noise spectrum extending to zero frequency at every accessible scale, attributable to the fundamental medium rather than to ordinary matter environments.

Prediction 11 — Detection Is Nucleation Kinetics

The relation. The detector click is a thermally activated topological event (kink-pair nucleation, FND-STRAND-006/007), committing to the full first-passage package where the model applies: near-exponential waiting times at fixed drive, Arrhenius dark counts, latency decreasing with channel energy. DERIVED: the event's existence, localization, irreversibility, and channel-energy drive (Arrhenius-linear in amplitude-squared, $r^2 = 0.984$). HONEST CAVEAT: parts of this package resemble known avalanche-detector phenomenology — the overlap is why the model is plausible, and the commitment is to the WHOLE package. Falsifier: confirmed click statistics that are non-activated and non-first-passage in a regime where the model's detector physics applies.

Prediction 12 — Spin-½ Requires Extension (one-sided)

The relation. The half-angle response is the SU(2) lift, which a point orientation cannot carry and a framed strand cannot avoid (FND-STRAND-005; measured at 10^{-9} for arbitrary axes and profiles). Commitment: every spin-½ particle has internal structure at SOME finite scale. Scale-open, hence one-sided in practice: compositeness bounds may rise indefinitely without contradiction, but a CONFIRMED structureless point with spin-½ at all scales would falsify the mechanism outright.

Prediction 13 — Tsirelson Forever, with the Failed Organ Named

The relation. CHSH-type correlations never exceed $2\sqrt{2}$ — shared with quantum mechanics; distinctive here is the WHY and therefore the autopsy: supra-Tsirelson statistics require a setting-dependence channel of polynomial degree greater than 24 where this framework's physics has degree exactly 1 (QB-018/019/022). Falsifier: any confirmed loophole-free violation above $2\sqrt{2}$ — which kills the measurement sector at a named claim, not merely embarrasses it.

Prediction 14 — Born's Square from Channel Energy

The relation. The quadratic ($p = 2$) response near detection threshold is the statement that nucleation is driven by channel ENERGY, quadratic in amplitude (QB-011 protocol; FND-STRAND-005 B4, $p = 1.99$). Falsifier: a confirmed, reproducible deviation from quadratic threshold response — breaking the chain at a specific link (the energy-drive identification), leaving the rest standing for inspection.

Prediction 15 — Analog Universality (testable now)

The relation. Any engineered system realizing the twist-chain physics — coupled pendulum arrays, magnetic domain-wall lattices, cold-atom sine-Gordon simulators — must exhibit the friction cliff (seven-order exponential mobility suppression, slope ≈ −8.5 per width unit, arrest/coast dichotomy) with the LAW transferring across implementations and only prefactors carrying the coupling convention (FND-KIN-002; FND-STRAND-002, slope agreement 0.16%). Engineered filament brushes should likewise show the transparent-to-solid coverage crossover at gap ≈ contact size with the measured inverse-fourth-power approach (FND-STRAND-004). No knowledge of the absolute strand scale is required, because the experimenter builds the system.

A retrodiction, declared as such: the Călugăreanu ledger and the supercoiling instability (FND-STRAND-003) are confirmed daily by DNA mechanics — consistency evidence for the strand engine, counted here as retrodiction, not prediction.

Honest Status of the Framework

For any reviewer evaluating these predictions, the following should be stated plainly:

The rope hypothesis is non-standard, self-published, and not peer-reviewed. These predictions are presented as empirical consequences of specific stated relations, not as derivations from established physics.

Two of the four testable numbers (the neutrino π/12 offset; the MOND 2π coefficient) rest on inputs that are empirical choices, not derived. They are falsifiable but not parameter-free, and we have marked exactly where.

The framework's deepest open problem — deriving absolute particle masses (and hence atomic and molecular masses, and the absolute strand scale that would close the scale-open entries above) — is UNSOLVED. A systematic search ruled out eight natural mechanisms; dimensional transmutation was closed by a structural result. The framework openly publishes this failure alongside its successes.

The strongest internal result — lepton mass RATIOS from $\varphi = (3+\Phi)\theta_W$ to ~0.5% — is a CONJECTURE, not a theorem, and is not listed among the numbers above.

Within the measurement sector, two grants remain and are named wherever they matter: the pair is one thing (Bell-protected), and the weave is warm (a temperature, not a law).

Acknowledgment of Origin

The core idea underlying this work — that physical phenomena arise from a two-strand rope-like structure connecting matter, with light as a transverse disturbance along the rope, and the principle that physical theories must describe objects rather than non-object concepts — originates with Bill Gaede (the "Rope Hypothesis" / "Thread Theory"). The predictions and the computational, field-theoretic, and topological development presented here are an independent, modified mathematization that departs from his formulation in substantive ways. Crediting this origin is not a claim that Gaede endorses this work, nor an endorsement of all of his positions; it is the honest acknowledgment of where the foundational idea came from.

References

Eskilt, J. R. & Komatsu, E. 2022, Phys. Rev. D 106, 063503 (cosmic birefringence).

Brannen, C. 2006, unpublished (Koide neutrino phase offset π/12).

Koide, Y. 1983, Phys. Lett. B 120, 161 (charged-lepton mass relation).

Lelli, F., McGaugh, S. & Schombert, J. 2016, AJ 152, 157 (SPARC database).

Software: rope-framework v2.2.5 (open source; DOI 10.5281/zenodo.21430784) — frozen benchmark IDs; every number in this note reproducible from the registry (claims.yaml) and its 155 passing benchmarks.

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Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth; every benchmark reruns from source.

• CHEM-GEO-002 [Derived]: HEAVY-HYDRIDE 90-DEGREE ASYMPTOTE [benchmark: benchmarks/em/heavy_hydride_asymptote.py]

• FND-STRAND-005 [Modeled]: THE INTERACTION MODEL ON LITERAL STRANDS — the framed strand is the SU(2) lift [benchmark: benchmarks/foundations/strand_interaction_model.py]

• FND-STRAND-006 [Modeled]: THE NUCLEATION EVENT ON STRANDS — the dot as topological event [benchmark: benchmarks/foundations/strand_nucleation.py]

• FND-STRAND-007 [Modeled]: THE WEAVE AS RESERVOIR — the bath derived [benchmark: benchmarks/foundations/strand_weave_reservoir.py]

• FND-STRAND-008 [Modeled]: THE WEAVE SPECTRUM FROM THE ENGINE — the bath is gapped [benchmark: benchmarks/foundations/strand_weave_spectrum.py]

• QB-019 [Derived]: MECHANISM-UNITY SELECTS TSIRELSON [benchmark: benchmarks/bell/tsirelson_selection.py]

• QB-022 [Modeled]: MECHANISM-UNITY DECOMPOSED AND GROUNDED (SOVO) [benchmark: benchmarks/bell/mechanism_unity_grounded.py]